Precalculus II

An Original Ten-Week Course Reader

Dr. Lee MATH& 142

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1 Angles, Rotation, and the Unit Circle

This week's central question: How can we turn rotation into coordinates? The answer begins with the natural geometry of a circle.

1.1 Angles measure turning

An angle records a rotation from one ray to another. Counterclockwise rotations are positive, and clockwise rotations are negative. A rotation can exceed one full revolution. For example, 450° and 90° have the same terminal ray.

Definition 1.1. Two angles are *coterminal* if they have the same initial and terminal rays. They differ by an integer multiple of 360°, or 2π radians.

Example 1.1. A positive coterminal angle for -50° is $-50^\circ + 360^\circ = 310^\circ$.

1.2 Why radians are natural

For a circle of radius r with intercepted arc length s, define the radian measure $\theta = s/r$. This is a ratio of lengths. A full circumference is $2\pi r$, so one revolution measures 2π radians:

$$\boxed{\theta = \frac{s}{r}}, \qquad 180^\circ = \pi \text{ radians}.$$

Example 1.2. Convert 150° to radians: $150(\pi/180) = 5\pi/6$. Convert $7\pi/4$ radians to degrees: $(7\pi/4)(180/\pi) = 315^\circ$.

Try it. Convert 225° to radians and $-\pi/3$ radians to degrees, and sketch their terminal rays.

1.3 Arc length and angular speed

Rearrange the definition of radian measure to obtain $s = r\theta$. If rotation through θ takes time t, angular speed is $\omega = \theta/t$ and tangential speed is $v = r\omega$.

Example 1.3. A wheel of radius 0.4 m turns through $3\pi/2$ radians. A point on its rim travels $s = 0.4(3\pi/2) = 0.6\pi$ m.

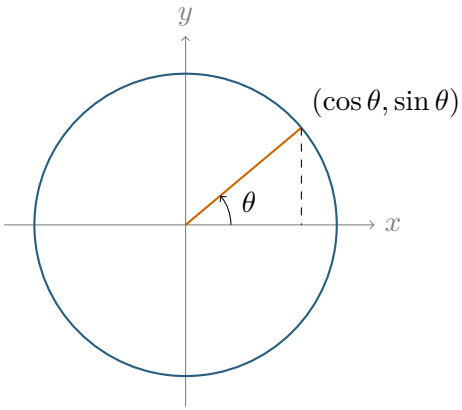
Try it. A Ferris wheel of radius 12 m makes one revolution every 40 s. Find its angular speed and a passenger's tangential speed.

1.4 Rotation becomes coordinates

Take a circle of radius 1 centered at the origin. The point at angle θ, measured from the positive x-axis, has coordinates

$$\boxed{(x, y) = (\cos\theta, \sin\theta)}.$$

Cosine is the horizontal coordinate and sine is the vertical coordinate. Because the point lies on $x^2 + y^2 = 1$, we get $\cos^2\theta + \sin^2\theta = 1$.



1.5 Reference angles and periodicity

Special angles arise from right triangles:

θ	$\cos\theta$	$\sin\theta$
0	1	0
$\pi/6$	$\sqrt{3}/2$	$1/2$
$\pi/4$	$\sqrt{2}/2$	$\sqrt{2}/2$
$\pi/3$	$1/2$	$\sqrt{3}/2$
$\pi/2$	0	1

The reference angle supplies the absolute values in every quadrant; the quadrant gives the signs.

Example 1.4. At $5\pi/6$, the reference angle is $\pi/6$ and the point is in quadrant II, so $\cos(5\pi/6) = -\sqrt{3}/2$ and $\sin(5\pi/6) = 1/2$.

After one full revolution the point returns to its previous position:

$$\sin(\theta + 2\pi) = \sin\theta, \qquad \cos(\theta + 2\pi) = \cos\theta.$$

This geometric periodicity will become the foundation of sinusoidal graphs.

1.6 Exercises

1. Convert $-135°$ and $300°$ to radians.
2. Convert $11\pi/6$ and $-5\pi/4$ to degrees.
3. Find an angle in $[0, 2\pi)$ coterminal with $19\pi/6$.
4. A circle of radius 8 cm subtends an angle of $5\pi/12$. Find the arc length.
5. A turntable makes $33\frac{1}{3}$ revolutions per minute. Find angular speed in radians per second.
6. Explain why sine and cosine must lie between -1 and 1.
7. Find unit-circle coordinates at $2\pi/3$, $5\pi/4$, and $11\pi/6$.
8. A Ferris wheel has radius 10 m and center $(0, 12)$. Write a rider's coordinates after turning through θ from the rightmost point.

Looking ahead. Next week we extend sine and cosine to the remaining four trigonometric functions.

2 The Six Trigonometric Functions

Central question. How do six functions arise naturally from the same moving point on a circle?

2.1 From a point to six functions

Last week we studied the unit-circle point $P = (x, y) = (\cos\theta, \sin\theta)$. The two coordinates determine four additional ratios whenever the denominators are nonzero:

$$\sin\theta = y, \quad \cos\theta = x, \quad \tan\theta = \frac{y}{x},$$
$$\csc\theta = \frac{1}{y}, \quad \sec\theta = \frac{1}{x}, \quad \cot\theta = \frac{x}{y}.$$

These are not six unrelated measurements. They are six different ways of reading the same geometric information.

Definition 2.1. The *tangent* function is $\tan\theta = \sin\theta/\cos\theta$ where $\cos\theta \neq 0$. The functions csc, sec, cot are the reciprocals of sin, cos, tan respectively, wherever defined.

Example 2.1. At $\theta = \pi/3$ the circle point is $(1/2, \sqrt{3}/2)$. Thus $\tan\theta = \sqrt{3}$, $\sec\theta = 2$, $\csc\theta = 2/\sqrt{3} = 2\sqrt{3}/3$, and $\cot\theta = 1/\sqrt{3} = \sqrt{3}/3$.

2.2 Signs and the quadrants

The coordinate signs determine every trig-function sign. In quadrant I both coordinates are positive; in II, only sine is positive; in III, tangent is positive because both coordinates are negative; and in IV, only cosine is positive. Reciprocal functions have the same signs as their corresponding original functions.

Example 2.2. For $\theta = 5\pi/4$, the reference angle is $\pi/4$ and $P = (-\sqrt{2}/2, -\sqrt{2}/2)$. Therefore $\tan\theta = 1$, $\sec\theta = -\sqrt{2}$, and $\csc\theta = -\sqrt{2}$.

2.3 Where are the functions undefined?

Division by zero has no meaning. In particular, $\tan\theta$ and $\sec\theta$ do not exist where $\cos\theta = 0$, at $\pi/2 + k\pi$. Similarly $\cot\theta$ and $\csc\theta$ do not exist where $\sin\theta = 0$, at $k\pi$, for any integer k. The word *undefined* matters: an undefined value is not the number zero.

Try it. Find all six trig-function values at $\theta = 3\pi/2$, identifying every undefined value.

2.4 Right triangles are a special case

For an acute angle θ in a right triangle, scale the triangle so its hypotenuse has length 1. The opposite and adjacent sides then give the coordinates $\sin\theta$ and $\cos\theta$. Restoring a general hypotenuse h gives

$$\sin\theta = \frac{\text{opposite}}{h}, \quad \cos\theta = \frac{\text{adjacent}}{h}, \quad \tan\theta = \frac{\text{opposite}}{\text{adjacent}}.$$

The unit circle is more general than the right-triangle definitions: it handles negative angles, obtuse angles, and angles exceeding a full revolution.

Example 2.3. A ramp rises 1.5 meters over a horizontal run of 6 meters. Its angle of inclination satisfies $\tan\theta = 1.5/6 = 1/4$. Therefore $\theta = \arctan(1/4) \approx 14.0°$.

2.5 Special angles without memorizing a disconnected table

An equilateral triangle of side 2 splits into two right triangles with sides $1, \sqrt{3}, 2$. This produces $\sin 30^\circ = 1/2$ and $\cos 30^\circ = \sqrt{3}/2$. A square of side 1 has diagonal $\sqrt{2}$, producing $\sin 45^\circ = \cos 45^\circ = \sqrt{2}/2$.



2.6 Applications: orientation and measurement

Suppose an observer stands 40 m from the foot of a vertical tower. The angle of elevation to its top is 35°. The tower's height h satisfies

$$\tan 35^\circ = \frac{h}{40}, \qquad h = 40 \tan 35^\circ \approx 28.0 \text{ m}.$$

Here the tangent function links an angle to a ratio of lengths. Always sketch the physical situation and label which distances are opposite and adjacent before calculating.

2.7 Exercises

1. At $\theta = 2\pi/3$, find all six functions exactly.
2. At $\theta = 7\pi/4$, find $\tan, \cot, \sec, \csc$.
3. Explain geometrically why $\tan(\pi/2)$ is undefined.
4. For which angles in $[0, 2\pi)$ is $\csc\theta$ undefined?
5. A right triangle has opposite side 5 and adjacent side 12. Find all six trig ratios exactly.
6. A ladder leans against a wall, reaching 8 m high. Its foot is 3 m from the wall. Find the angle with the ground.
7. A drone is viewed at an elevation angle of 28° from a point 75 m horizontally from the point below it. Estimate its altitude.
8. Show that $\sin^2\theta + \cos^2\theta = 1$ implies $1 + \tan^2\theta = \sec^2\theta$ wherever $\cos\theta \neq 0$.

3 Periodic Functions and Their Graphs

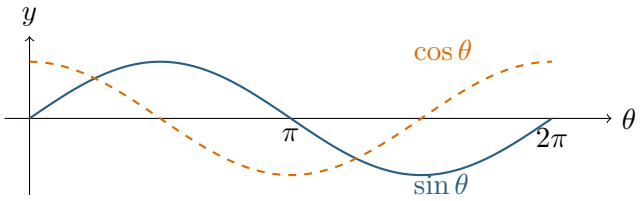
Central question. How does rotation become a repeating signal, and how can we recover the motion from a graph?

3.1 The circle makes a wave

For a point traveling around the unit circle with angle θ, its vertical coordinate is $y = \sin\theta$ and its horizontal coordinate is $x = \cos\theta$. Each revolution restores the same coordinates. Hence

$$\sin(\theta + 2\pi) = \sin\theta, \qquad \cos(\theta + 2\pi) = \cos\theta.$$

The *period* is the smallest positive input change that repeats the full function. Both sine and cosine have period 2π. Sine starts at 0 and initially rises; cosine starts at 1.



3.2 Amplitude and vertical translation

Consider $y = A\sin x + D$. Its midline is $y = D$, its amplitude is $|A|$, and its range is $[D - |A|, D + |A|]$. A negative A reflects the wave across the midline.

Example 3.1. The function $y = 3\sin x - 2$ has amplitude 3, midline $y = -2$, maximum 1, and minimum -5.

3.3 Period and horizontal scaling

For $y = \sin(Bx)$ with $B \neq 0$, one full cycle occurs when Bx changes by 2π; the period is $T = 2\pi/|B|$.

Example 3.2. The graph of $y = \cos(4x)$ repeats every $\pi/2$ units of input.

When x denotes time measured in seconds, the quantity $|B|$ is angular frequency in radians per second, and ordinary frequency is $f = 1/T = |B|/(2\pi)$ cycles per second.

3.4 Phase shift

A useful general form is

$$\boxed{y = A\sin(B(x - C)) + D}.$$

Here C is the horizontal shift, D the midline, $|A|$ the amplitude, and $2\pi/|B|$ the period. We should distinguish $B(x - C)$ from $Bx - C$: factoring first prevents an incorrect phase shift.

Example 3.3. For $y = 2\sin(3x - \pi) + 1 = 2\sin(3(x - \pi/3)) + 1$, the period is $2\pi/3$, the shift is $\pi/3$ to the right, the amplitude is 2, and the midline is 1.

3.5 Five landmark points

Rather than sketching a sinusoid by guesswork, divide one period into four equal intervals. For $A > 0$, sine begins on the midline, rises to a maximum after one quarter-period, returns to the midline, reaches a minimum, then returns to the midline. A negative amplitude reverses the rises and falls.

Try it. Find five landmark points of $y = -2\cos(\pi(x-1)) + 3$, showing one entire period.

3.6 Models of repeating motion

A Ferris wheel has radius 9 m, its center is 11 m above ground, and it makes one revolution every 30 s. If a passenger begins at the rightmost position and travels counterclockwise, their height is

$$h(t) = 11 + 9\sin\left(\frac{2\pi}{30}t\right).$$

Its amplitude is 9 m, midline is 11 m, and period is 30 s. A sinusoidal model can describe a wheel, a spring, a tide, or a musical tone, provided its parameters actually correspond to the chosen physical quantities.

3.7 Tangent and reciprocal graphs

The tangent function repeats after π, not 2π, because both sine and cosine change sign after a half turn and their ratio remains the same:

$$\tan(x + \pi) = \tan x.$$

Tangent has vertical asymptotes at $x = \pi/2 + k\pi$. The secant and cosecant graphs arise as reciprocals of cosine and sine; their breaks occur wherever the respective denominator is zero.

3.8 Exercises

1. Find the amplitude, midline, period, and range of $y = -4\sin(2x) + 1$.

2. Rewrite $y = 3\cos(5x - \pi) + 2$ in shifted form and identify the phase shift.

3. Give five landmark points for $y = 2\sin(x - \pi/2)$.

4. Explain why $y = \sin(2x)$ completes two periods on $[0, 2\pi]$.

5. A wheel of radius 6 m rotates once every 24 seconds. Its center is 8 m above the ground. Model the rider's height when starting at the rightmost point.

6. Identify the vertical asymptotes of $y = \tan(2x)$ on $[0, 2\pi)$.

7. Sketch one period of $y = 2\sec x$ and indicate all discontinuities.

8. Explain how you can determine amplitude and midline directly from observed maximum and minimum values.

4 Inverse Trigonometry and Identities

Central question: How can we recover an angle from its trigonometric value, and how do we prove two expressions represent the same function?

4.1 An inverse is not a reciprocal

The notation $\sin^{-1} x$ means the inverse function $\arcsin x$. It does not mean $1/\sin x = \csc x$. The inverse answers an angle question; the reciprocal answers a ratio question.

Sine is periodic, so an inverse function requires choosing an interval on which sine is one-to-one. We use $[-\pi/2, \pi/2]$. In exactly the same way, cosine uses $[0, \pi]$ and tangent uses $(-\pi/2, \pi/2)$.

$f(x)$	Domain	Range
$\arcsin x$	$[-1, 1]$	$[-\pi/2, \pi/2]$
$\arccos x$	$[-1, 1]$	$[0, \pi]$
$\arctan x$	$\mathbb{R}$	$(-\pi/2, \pi/2)$

Example 4.1. $\arcsin(1/2) = \pi/6$, $\arccos(-1/2) = 2\pi/3$, and $\arctan(-1) = -\pi/4$.

Example 4.2. $\arcsin(\sin(5\pi/6)) = \pi/6$, not $5\pi/6$, because the inverse reports its principal value.

4.2 Compositions and triangle geometry

The identity $\sin(\arcsin x) = x$ holds for $-1 \le x \le 1$, but $\arcsin(\sin\theta) = \theta$ only for θ in the principal range. Suppose $\theta = \arctan(3/4)$. Then a right triangle has legs $3, 4$ and hypotenuse 5, so

$$\sin(\arctan(3/4)) = \frac{3}{5}.$$

4.3 An identity is true throughout its domain

From the unit-circle equation we obtain

$$\boxed{\sin^2 x + \cos^2 x = 1}.$$

Dividing by $\cos^2 x$ (only where $\cos x \neq 0$) gives $1 + \tan^2 x = \sec^2 x$. Dividing by $\sin^2 x$ gives $1 + \cot^2 x = \csc^2 x$ where $\sin x \neq 0$.

Example 4.3. For $\sin x \neq 0$,

$$\frac{1 - \cos^2 x}{\sin x} = \frac{\sin^2 x}{\sin x} = \sin x.$$

This is an algebraic proof on the shared domain, not a numerical check at a few inputs.

4.4 Angle addition and double angles

Two successive plane rotations produce the sum identities

$$\cos(\alpha + \beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta,$$
$$\sin(\alpha + \beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta.$$

Set $\beta = -\gamma$ for difference identities and $\beta = \alpha$ for double-angle identities:

$$\sin(2x) = 2\sin x\cos x, \qquad \cos(2x) = \cos^2 x - \sin^2 x = 1 - 2\sin^2 x.$$

Example 4.4. $\sin(75^\circ) = \sin(45^\circ + 30^\circ) = (\sqrt{6} + \sqrt{2})/4$.

4.5 Applications: measured angles

An observation of a tower from 40 m away gives $\tan\theta = 3/4$. The elevation angle is $\arctan(3/4) \approx 36.9°$, while the height is exactly $40(3/4) = 30$ m. Inverse functions recover angles; ratios can often recover distances directly.

4.6 Exercises

1. Evaluate exactly $\arcsin(-\sqrt{2}/2)$ and $\arctan(\sqrt{3})$.
2. Explain why $\arcsin(\sin(3\pi/4)) \neq 3\pi/4$.
3. Find $\cos(\arcsin(5/13))$ with a triangle.
4. Find $\sin(\arctan(-2))$ exactly, including its sign.
5. Prove $(\sec^2 x - 1)/\tan x = \tan x$ on its domain.
6. Prove $(\sin x + \cos x)^2 = 1 + \sin(2x)$.
7. Calculate $\cos(75°)$ exactly by angle addition.
8. Express $\cos(2x)$ using $\sin x$ alone.

5 Trigonometric Equations and Applications

Central question. When a trigonometric function has a prescribed value, how can we find *all* the angles that produce it? How can we use angles to solve triangles that are not right triangles?

5.1 A first equation on the unit circle

Consider $\sin x = 1/2$. The unit circle has two points whose vertical coordinate is $1/2$, at $x = \pi/6$ and $x = 5\pi/6$. Every full revolution reproduces these points:

$$\boxed{x = \frac{\pi}{6} + 2\pi k \quad \text{or} \quad x = \frac{5\pi}{6} + 2\pi k, \qquad k \in \mathbb{Z}.}$$

An inverse trigonometric function gives one principal angle, not necessarily every solution. The unit circle and periodicity give the complete answer.

Example 5.1. Solve $\cos x = -\sqrt{2}/2$ for $0 \le x < 2\pi$. The solutions are $3\pi/4$ and $5\pi/4$.

5.2 Equations with transformations

To solve $\sin(2x) = \sqrt{3}/2$ on $[0, 2\pi)$, first solve for $u = 2x$ on $[0, 4\pi)$. We obtain $u = \pi/3, 2\pi/3, 7\pi/3, 8\pi/3$ and therefore

$$x = \frac{\pi}{6}, \frac{\pi}{3}, \frac{7\pi}{6}, \frac{4\pi}{3}.$$

A changed frequency changes how many full cycles fit into the requested interval.

Example 5.2. Solve $2\cos x - 1 = 0$ on $[0, 2\pi)$. Isolating cosine gives $\cos x = 1/2$, hence $x = \pi/3$ or $5\pi/3$.

5.3 Factoring and checking domains

An equation may contain more than one trig function. Algebraic techniques still work:

$$2\sin^2 x - \sin x = 0 \quad \Longrightarrow \quad \sin x(2\sin x - 1) = 0.$$

On $[0, 2\pi)$ we obtain $x = 0, \pi, \pi/6, 5\pi/6$. Dividing by $\sin x$ at the outset would have lost two solutions.

5.4 Trigonometric identities in equations

For example, $2\cos^2 x - 1 = 0$ becomes $\cos^2 x = 1/2$, hence $\cos x = \pm\sqrt{2}/2$. There are four solutions on $[0, 2\pi)$. Using identities can make equations simpler, but each transformation should preserve the original domain.

5.5 Solving triangles: notation

Write the vertices of a triangle as A, B, C, and the opposite side lengths as a, b, c. Its angles satisfy $A + B + C = \pi$. For right triangles, the six trig ratios are enough. For oblique triangles, two laws extend our tools.

Definition 5.1 (Law of Sines)**.** For every nondegenerate triangle,

$$\boxed{\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}.}$$

Example 5.3. A triangle has $A = 40°$, $B = 65°$, and $a = 9$ cm. Since $C = 75°$, the Law of Sines gives $b = 9\sin 65°/\sin 40° \approx 12.69$ cm.

5.6 The ambiguous SSA case

Given two sides and a nonincluded angle (SSA), the Law of Sines may produce zero, one, or two triangles. If $\sin B = s$, both $B = \arcsin s$ and $B = \pi - \arcsin s$ must be considered, and each candidate must fit with the known angle A in a triangle.

Example 5.4. Suppose $A = 30^\circ$, $a = 5$ and $b = 8$. Then $\sin B = 8 \sin 30^\circ / 5 = 0.8$. Both $B \approx 53.13^\circ$ and $B \approx 126.87^\circ$ are consistent with $A + B < 180^\circ$, so two distinct triangles exist.

Definition 5.2 (Law of Cosines)**.** For a triangle with sides a, b, c opposite angles A, B, C,

$$\boxed{c^2 = a^2 + b^2 - 2ab\cos C.}$$

When $C = 90^\circ$ this becomes the Pythagorean theorem. The cosine term measures how the included angle modifies the length of the opposite side.

Example 5.5. Two sides of a triangle are 7 m and 10 m, with included angle 60°. The opposite side satisfies $c^2 = 49 + 100 - 140(1/2) = 79$, so $c = \sqrt{79}$ m.

5.7 Applied modeling and interpretation

Two sensors observe a distant object from different locations. The baseline and their sight angles make a triangle. The Law of Sines can recover an inaccessible distance if we know a side and two angles. Always include the diagram, angle units, and a reasonableness check.

5.8 Exercises

1. Solve $\sin x = -\sqrt{3}/2$ for $0 \le x < 2\pi$.
2. Solve $2\sin(3x) = 1$ for $0 \le x < 2\pi$; find every solution.
3. Solve $2\cos^2 x - \cos x = 0$ on $[0, 2\pi)$.
4. Solve $\tan x = -1$ for all real x.
5. A triangle has $A = 35^\circ$, $B = 80^\circ$ and $a = 12$. Find C and b.
6. For $A = 30^\circ$, $a = 5$, $b = 8$, calculate both possible values of C in the SSA case.
7. Find the third side of a triangle when two sides have length 6 and 9 and their included angle is 120°.
8. Explain why the Law of Cosines becomes the Pythagorean theorem for right triangles.

6 Vectors and Geometric Applications

Central question. How do we describe a quantity that has both a size and a direction?

6.1 From coordinates to directed displacements

In the first half of the course we used $(\cos\theta, \sin\theta)$ to locate a point on a circle. A vector uses the same coordinates to describe a *displacement*: how far to move horizontally and vertically. Write $\mathbf{v} = \langle a, b\rangle$. Its magnitude is

$$\boxed{\|\mathbf{v}\| = \sqrt{a^2 + b^2}}.$$

Vectors with the same components represent the same displacement even when drawn at different starting points.

Example 6.1. A walker moves 3 km east and 4 km north. The resulting vector is $\langle 3, 4\rangle$ km and its magnitude is 5 km. The route traveled is 7 km, while the displacement is only 5 km.

6.2 Vector arithmetic

Addition combines displacements:

$$\langle a, b\rangle + \langle c, d\rangle = \langle a + c, b + d\rangle.$$

Scalar multiplication changes their length, and a negative scalar reverses direction:

$$k\langle a, b\rangle = \langle ka, kb\rangle.$$

Subtracting vectors compares locations. If $P = (p_1, p_2)$ and $Q = (q_1, q_2)$, then $\overrightarrow{PQ} = \langle q_1 - p_1, q_2 - p_2\rangle$.

Example 6.2. For $\mathbf{u} = \langle 2, -1\rangle$ and $\mathbf{v} = \langle -3, 4\rangle$, we have $\mathbf{u} + \mathbf{v} = \langle -1, 3\rangle$ and $2\mathbf{u} - \mathbf{v} = \langle 7, -6\rangle$.

6.3 Direction and trigonometry

A unit vector has magnitude 1. The unit vector pointing at angle θ is

$$\mathbf{e}_\theta = \langle \cos\theta, \sin\theta\rangle.$$

A vector of magnitude r in that direction is

$$\boxed{\mathbf{v} = r\langle \cos\theta, \sin\theta\rangle}.$$

Conversely, for a nonzero vector $\langle a, b\rangle$, the angle must be chosen in the appropriate quadrant. An inverse tangent applied only to b/a may miss that quadrant.

Example 6.3. A force of 20 N directed 30° above the horizontal has components $\langle 10\sqrt{3}, 10\rangle$ N.

6.4 The dot product measures alignment

Define

$$\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2.$$

The geometric interpretation is

$$\boxed{\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\|\|\mathbf{v}\| \cos\phi},$$

where ϕ is the angle between nonzero vectors. Positive dot product means an acute angle; negative means an obtuse angle; zero means perpendicular vectors.

Example 6.4. With $\mathbf{u} = \langle 3, 4\rangle$ and $\mathbf{v} = \langle 4, -3\rangle$, the dot product is $12 - 12 = 0$. Both vectors are nonzero, so they are perpendicular.

6.5 Projection and useful components

The scalar component of $\mathbf{u}$ in the direction of a nonzero vector $\mathbf{v}$ is

$$\operatorname{comp}_{\mathbf{v}} \mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{v}\|}.$$

The corresponding vector projection is

$$\operatorname{proj}_{\mathbf{v}} \mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{v}\|^2}\mathbf{v}.$$

Projection is the part of one vector aligned with another. In physics, the work done by a constant force $\mathbf{F}$ across displacement $\mathbf{d}$ is $W = \mathbf{F} \cdot \mathbf{d}$, measured in joules when force is in newtons and displacement in meters.

Example 6.5. A force $\langle 10, 5\rangle$ N acts during displacement $\langle 3, 0\rangle$ m. The work is 30 J: only the horizontal force component contributes.

6.6 A geometric connection to the Law of Cosines

For $\mathbf{u}, \mathbf{v}$ we have

$$\|\mathbf{u} - \mathbf{v}\|^2 = (\mathbf{u} - \mathbf{v}) \cdot (\mathbf{u} - \mathbf{v}) = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - 2\|\mathbf{u}\|\|\mathbf{v}\| \cos \phi.$$

Thus the Law of Cosines from Module 5 is also a theorem about vector differences.

6.7 Exercises

1. Find the vector from $P = (-2, 3)$ to $Q = (5, -1)$ and its magnitude.
2. For $\mathbf{u} = \langle 2, 5\rangle$, $\mathbf{v} = \langle -1, 3\rangle$, compute $\mathbf{u} + \mathbf{v}$ and $3\mathbf{u} - 2\mathbf{v}$.
3. Find exact components of a displacement of magnitude 12 at angle 150°.
4. Find a unit vector in the direction of $\langle -3, 4\rangle$.
5. Compute $\langle 2, -1\rangle \cdot \langle 5, 3\rangle$ and interpret its sign.
6. Determine whether $\langle 6, 2\rangle$ and $\langle -1, 3\rangle$ are perpendicular.
7. Find $\operatorname{proj}_{\mathbf{v}} \mathbf{u}$ when $\mathbf{u} = \langle 4, 3\rangle$ and $\mathbf{v} = \langle 1, 0\rangle$.
8. A force of 50 N acts 60° from the direction of a 4 m displacement. Calculate the work done.

7 Polar Coordinates and Complex Numbers

Central question. What happens when we describe a point by its distance and direction instead of its horizontal and vertical coordinates?

7.1 Two descriptions of the same point

Cartesian coordinates specify (x, y). Polar coordinates specify (r, θ), where r is the directed radial distance and θ is an angle from the positive horizontal axis:

$$\boxed{x = r\cos\theta, \qquad y = r\sin\theta.}$$

For $r \geq 0$, the distance is $r = \sqrt{x^2 + y^2}$. The angle must be chosen using the signs of both coordinates, not just $\arctan(y/x)$.

Example 7.1. The polar point $(4, 5\pi/6)$ corresponds to $(x, y) = (-2\sqrt{3}, 2)$. Its distance from the origin is 4, and it lies in Quadrant II.

Try it. Convert $(r, \theta) = (6, -\pi/3)$ to Cartesian coordinates. Then describe its direction using an angle in $[0, 2\pi)$.

7.2 Polar coordinates are not unique

Angles that differ by 2π represent the same direction, so (r, θ) and $(r, \theta + 2\pi k)$ describe the same point. A negative radius reverses the ray:

$$(r, \theta) = (-r, \theta + \pi).$$

For example, $(2, \pi/3)$, $(2, 7\pi/3)$, and $(-2, 4\pi/3)$ are three descriptions of the same point. At the origin $r = 0$, every angle describes the same location.

7.3 Complex numbers as plane vectors

Introduce i satisfying $i^2 = -1$. A complex number $z = a + bi$ corresponds to the point (a, b) in the complex plane. Its modulus and argument are

$$|z| = \sqrt{a^2 + b^2}, \qquad \arg z = \theta \quad \text{when } z = |z|(\cos\theta + i\sin\theta).$$

The argument of $z = 0$ is undefined. As with polar angles, nonzero arguments are only unique after choosing a branch, such as $(-\pi, \pi]$.

Example 7.2. For $z = -\sqrt{3} + i$, the modulus is 2 and one argument is $5\pi/6$. Thus

$$z = 2\left(\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6}\right).$$

7.4 Addition and multiplication mean different geometries

Complex addition follows the vector rule:

$$(a + bi) + (c + di) = (a + c) + (b + d)i.$$

Complex multiplication uses $i^2 = -1$:

$$(a + bi)(c + di) = (ac - bd) + (ad + bc)i.$$

If $z = r(\cos\alpha + i\sin\alpha)$ and $w = s(\cos\beta + i\sin\beta)$, the angle-addition formulas give

$$\boxed{zw = rs(\cos(\alpha+\beta) + i\sin(\alpha+\beta)).}$$

Therefore multiplying complex numbers multiplies distances from the origin and *adds angles*. A unit complex number acts as a rotation. This unifies the unit circle from Module 1 with vector geometry from Module 6.

Example 7.3. Multiplying $2+i$ by i gives $i(2+i) = -1+2i$. The vector $(2, 1)$ rotates counterclockwise by $90°$ to $(-1, 2)$, without changing its length.

7.5 De Moivre's theorem and roots

For integer n, repeated multiplication gives

$$\boxed{[r(\cos\theta + i\sin\theta)]^n = r^n(\cos n\theta + i\sin n\theta)}$$

whenever the expression makes sense; for negative n, require $r > 0$. For example, $(\cos(\pi/6) + i\sin(\pi/6))^3 = i$. To solve $z^n = R(\cos\phi + i\sin\phi)$ with $R > 0$, take

$$z_k = R^{1/n}\left(\cos\frac{\phi+2\pi k}{n} + i\sin\frac{\phi+2\pi k}{n}\right), \qquad k = 0, \ldots, n-1.$$

These n distinct roots are equally spaced around a circle.

Example 7.4. The four fourth roots of 1 are $1, i, -1, -i$. They are separated by a quarter-turn.

7.6 Polar graphs: a different way to draw a curve

In a polar equation such as $r = 2\cos\theta$, the radius changes with the angle. Multiply by r: $r^2 = 2r\cos\theta$, or $x^2 + y^2 = 2x$. Completing the square yields

$$(x-1)^2 + y^2 = 1.$$

Thus a simple polar equation can represent an ordinary circle displaced from the origin.

7.7 Exercises

1. Convert $(5, 2\pi/3)$ to Cartesian coordinates.
2. Give two distinct polar coordinate representations of $(\sqrt{2}, \sqrt{2})$.
3. Write $-1 - \sqrt{3}i$ in polar form with argument in $[0, 2\pi)$.
4. Calculate $(2+i)(1-3i)$ in rectangular form.
5. Interpret multiplication by $-i$ as a plane transformation.
6. Evaluate $(\cos(\pi/4) + i\sin(\pi/4))^4$.
7. Find all three cube roots of 1 and plot them on the unit circle.
8. Convert the polar curve $r = 4\sin\theta$ into a Cartesian equation and identify the curve.

8 Parametric Equations and Conic Sections

Central question. How can the same curve be described by coordinates, by a moving point, or by a geometric rule?

8.1 A moving point traces a curve

In a parametric description, both coordinates depend on a parameter t:

$$\boxed{x = f(t), \qquad y = g(t).}$$

Think of t as time when a particle moves. The curve records its location, while the parameterization also records direction and timing.

Example 8.1. For $x = 2t + 1$, $y = t - 3$, eliminating $t = (x - 1)/2$ gives $y = (x - 7)/2$. If t increases, the point moves up and right along this line.

8.2 Eliminating the parameter

For $x = 3\cos t$, $y = 2\sin t$, square the normalized equations and add:

$$\boxed{\frac{x^2}{9} + \frac{y^2}{4} = 1.}$$

The equation gives the whole ellipse when t ranges over $[0, 2\pi]$. A restricted interval, such as $[0, \pi]$, traces only its upper half. Different parameterizations may have the same image but different speeds or directions.

Example 8.2. Both $(x, y) = (\cos t, \sin t)$ and $(x, y) = (\cos 2t, \sin 2t)$ trace the unit circle for $0 \le t \le 2\pi$; the second makes two revolutions.

8.3 Projectile motion as a parametric curve

Under constant downward acceleration g, a projectile launched from (x_0, y_0) at speed v_0 and angle θ satisfies

$$x(t) = x_0 + v_0 \cos\theta\, t, \qquad y(t) = y_0 + v_0 \sin\theta\, t - \tfrac{1}{2} g t^2.$$

When the horizontal velocity is nonzero, eliminating t gives a parabola. The motion model adds information the Cartesian curve alone does not supply, including direction and location at a given time.

8.4 Conics: circles and ellipses

A *circle* is the set of points at a fixed distance r from a center (h, k):

$$(x - h)^2 + (y - k)^2 = r^2.$$

An *ellipse* can be defined by a constant sum of distances to two foci. In standard form, for $a \ge b > 0$,

$$\frac{(x - h)^2}{a^2} + \frac{(y - k)^2}{b^2} = 1.$$

Its horizontal semiaxis is a and vertical semiaxis is b; when the major axis is horizontal, focal distance obeys $c^2 = a^2 - b^2$.

Example 8.3. For $x^2/25 + y^2/9 = 1$, the vertices are $(\pm 5, 0)$, and the foci are $(\pm 4, 0)$.

8.5 Parabolas and their focus

A *parabola* is the set of points equidistant from a fixed point (focus) and a fixed line (directrix). The equation

$$(x-h)^2 = 4p(y-k)$$

has vertex (h,k), focus $(h,k+p)$, and directrix $y=k-p$. Its axis is vertical; $p>0$ opens upward and $p<0$ downward.

Example 8.4. For $x^2=8y$, we have $4p=8$, so the focus is $(0,2)$ and the directrix is $y=-2$.

8.6 Hyperbolas and differences of distances

A *hyperbola* consists of points for which the absolute difference of distances to two foci is fixed. A horizontal hyperbola has standard form

$$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1.$$

Its vertices are $(h\pm a,k)$, its foci are $(h\pm c,k)$ where $c^2=a^2+b^2$, and its asymptotes are $y-k=\pm(b/a)(x-h)$.

Example 8.5. The hyperbola $x^2/9 - y^2/16 = 1$ has vertices $(\pm 3,0)$, foci $(\pm 5,0)$, and asymptotes $y=\pm 4x/3$.

8.7 Changing descriptions

A parametric curve may be easier to generate than to recognize from a Cartesian equation. A Cartesian equation reveals symmetry or an axis. A focal definition explains why the curve arises in optics, navigation, and geometry. None of these descriptions replaces the others; each answers a different question.

8.8 Exercises

1. Eliminate t from $x=t+2$, $y=3t-1$ and describe the motion as t increases.
2. Find the Cartesian equation traced by $x=4\cos t$, $y=3\sin t$.
3. Explain how $(\cos t,\sin t)$ and $(\cos 3t,\sin 3t)$ differ as parameterizations.
4. A projectile leaves the origin at 20 m/s at 30°, with $g=9.8$ m/s^2. Write its parametric equations.
5. Identify center and radius of $(x-2)^2+(y+1)^2=16$.
6. Find the foci of $x^2/36+y^2/20=1$.
7. Find the focus and directrix of $(x-1)^2=12(y+2)$.
8. Find the vertices and asymptotes of $x^2/4-y^2/9=1$.

9 Sequences and Series

Central question. How do repeated steps produce patterns, and how can we predict both a later value and a cumulative total?

9.1 A sequence is an ordered list

A sequence assigns a number a_n to each positive integer n. For example, $a_n = 3 + 2(n-1)$ produces $3, 5, 7, 9, \ldots$. We distinguish the *term* a_n from a *partial sum* $S_n = a_1 + \cdots + a_n$.

9.2 Arithmetic sequences: repeated addition

An arithmetic sequence has a fixed common difference d:

$$a_{n+1} = a_n + d, \qquad \boxed{a_n = a_1 + (n-1)d}.$$

Example 9.1. A theater's first row has 12 seats, with two more seats in each successive row. Row 15 has $12 + 14(2) = 40$ seats.

Adding the first and last terms in reverse order gives

$$\boxed{S_n = \frac{n}{2}(a_1 + a_n) = \frac{n}{2}(2a_1 + (n-1)d)}.$$

9.3 Geometric sequences: repeated multiplication

A geometric sequence has a fixed common ratio q:

$$a_{n+1} = qa_n, \qquad \boxed{a_n = a_1 q^{n-1}}.$$

When $q > 1$, positive terms grow exponentially. When $0 < q < 1$, they decay toward zero.

Example 9.2. A population of 500 organisms increases by 8 percent each generation. The model is $a_n = 500(1.08)^{n-1}$ if generation 1 is the initial observation.

For $q \neq 1$, multiply the finite sum by q and subtract:

$$S_n = a_1 + a_1 q + \cdots + a_1 q^{n-1}, \qquad \boxed{S_n = a_1 \frac{1 - q^n}{1 - q}}.$$

If $q = 1$, the sum is na_1. The formulas refer to *finite* sums and work for ratios outside the interval $(0, 1)$ as well.

9.4 Sigma notation

The compact expression $\sum_{k=1}^{n} a_k$ means add the terms for integer indices $k = 1, \ldots, n$. For instance,

$$\sum_{k=1}^{4} (2k + 1) = 3 + 5 + 7 + 9 = 24.$$

The symbol $\sum$ tells us to add, while its bounds tell us which indices are included.

9.5 Infinite geometric series

If $|q| < 1$, then $q^n \to 0$. The finite geometric sum approaches

$$\boxed{\sum_{n=0}^{\infty} aq^n = \frac{a}{1-q}}, \qquad |q| < 1.$$

The condition matters: the series $1 + 2 + 4 + \cdots$ does not approach a finite sum.

Example 9.3. A ball bounces to 60% of its previous height. Released from height 5 m, the total vertical travel after infinitely many idealized bounces is $5+2(5)(0.6+0.6^2+\cdots) = 5+10(0.6/0.4) = 20$ m.

9.6 Recursive rules beyond arithmetic and geometric patterns

Not every sequence is arithmetic or geometric. Consider

$$a_{n+1} = \frac{a_n + 6}{2}, \qquad a_1 = 0.$$

The first terms are $0, 3, 4.5, 5.25, \ldots$. Let $e_n = a_n - 6$ denote the difference from the fixed point. Then $e_{n+1} = e_n/2$, so $e_n = -6(1/2)^{n-1}$ and

$$a_n = 6 - 6(1/2)^{n-1}.$$

The original recursion is not geometric, but its *error from equilibrium* is. This is a useful bridge to feedback and dynamical models.

9.7 Exercises

1. Write the first five terms of $a_n = 7 - 3(n-1)$ and find a_{20}.
2. An arithmetic sequence has $a_1 = 4$, $d = 5$. Calculate S_{30}.
3. A geometric sequence has $a_1 = 81$, $q = 1/3$. Find a_6.
4. Compute $\sum_{k=0}^{5} 2(3/2)^k$ exactly.
5. Explain the difference between $\sum_{k=1}^{n} k$ and its final term n.
6. Determine whether $5 + 5/2 + 5/4 + \cdots$ converges and find its sum.
7. A machine loses 12 percent of its value per year. Express its value after n years if it initially costs \$12,000.
8. For $a_{n+1} = (a_n + 10)/2$, $a_1 = 2$, find the fixed point and an explicit formula for a_n.

10 Synthesis: Motion and Patterns

Central question. How can the different mathematical languages of this course describe one evolving system?

10.1 One motion, several descriptions

Imagine a point traveling counterclockwise around a circle of radius R with angular speed ω. Its angle at time t is $\theta(t) = \theta_0 + \omega t$. In parametric form,

$$x(t) = R\cos(\theta_0 + \omega t), \qquad y(t) = R\sin(\theta_0 + \omega t).$$

Its polar form is $(r, \theta) = (R, \theta_0 + \omega t)$, and its complex form is

$$\boxed{z(t) = R\big(\cos(\theta_0 + \omega t) + i\sin(\theta_0 + \omega t)\big)}.$$

These are not competing models: they encode the same physical position in different coordinate languages.

10.2 Velocity, vectors, and periodicity

Over a short time interval Δt, the displacement vector is

$$\Delta\mathbf{r} = \langle x(t + \Delta t) - x(t), y(t + \Delta t) - y(t)\rangle.$$

For constant angular speed the point completes a revolution after $T = 2\pi/|\omega|$ when $\omega \neq 0$. Its horizontal and vertical coordinates repeat with that period. The direction of motion is tangent to the circle and perpendicular to the radius: this is the geometric heart of circular motion.

10.3 Sampling creates a sequence

Suppose we record the position every Δt seconds. The resulting sequence of complex positions is

$$z_n = R\big(\cos(\theta_0 + n\delta) + i\sin(\theta_0 + n\delta)\big), \qquad \delta = \omega\Delta t.$$

Writing $u = \cos\delta + i\sin\delta$ gives

$$\boxed{z_{n+1} = uz_n, \qquad z_n = u^n z_0.}$$

De Moivre's theorem then explains every sample as a repeated rotation. If $\delta = 2\pi/m$ for a positive integer m, the sampled orbit closes after m steps (provided this is the smallest such positive step count). If $\delta/(2\pi)$ is irrational, no two different samples have exactly the same position.

10.4 Scaling and rotation together

Now consider the recurrence $z_{n+1} = qz_n$ where

$$q = \rho(\cos\delta + i\sin\delta).$$

Its explicit solution is $z_n = q^n z_0$. Consequently,

$$|z_n| = \rho^n|z_0|, \qquad \arg z_n \equiv \arg z_0 + n\delta \pmod{2\pi}$$

for nonzero terms. If $0 < \rho < 1$, the points spiral toward the origin; if $\rho > 1$, they spiral outward; and if $\rho = 1$, they remain on a circle. Thus a geometric sequence and a rotation combine to produce a spiral.

10.5 Sums represent accumulated effects

If a system moves by step vectors $\mathbf{v}_0, \mathbf{v}_1, \ldots$, its total displacement after N steps is the vector sum $\sum_{n=0}^{N-1} \mathbf{v}_n$. For a complex-valued geometric pattern $\Delta z_n = aq^n$ with $q \neq 1$,

$$z_N - z_0 = \sum_{n=0}^{N-1} aq^n = a\frac{1-q^N}{1-q}.$$

Notice that the location sequence z_n and the step sequence Δz_n are different objects. Careful modeling starts by distinguishing what is being measured.

10.6 Different curves arise from different constraints

Circular and elliptical motion can be expressed parametrically. A projectile follows a parabola when horizontal velocity is constant and vertical acceleration is constant. Polar formulas express rays and radial patterns directly. The useful representation depends on whether we need a shape, a position, a direction, or a rule for change.

10.7 A complete example

A sensor begins at $z_0 = 2$ in the complex plane. Every second its controller multiplies the position by $q = \frac{1}{2}(\cos 60^\circ + i \sin 60^\circ)$. After n steps,

$$z_n = 2\left(\frac{1}{2}\right)^n (\cos(60^\circ n) + i \sin(60^\circ n)).$$

After three steps the modulus is $1/4$, and the direction is 180°. Hence $z_3 = -1/4$. The modulus obeys a geometric sequence, the direction obeys an arithmetic progression of angles, and the position follows a discrete inward spiral.

10.8 Exercises

1. Write Cartesian, polar, and complex formulas for a circle of radius 5 rotating once every 20 seconds, starting on the positive horizontal axis.

2. Sample that motion every five seconds. Give its first five complex positions and identify its repetition period in samples.

3. For $z_{n+1} = iz_n$ and $z_0 = 1 + i$, calculate $z_1, \ldots, z_4$ and describe the geometry.

4. For $z_{n+1} = (1/2)iz_n$, $z_0 = 4$, find z_4 and explain the change in magnitude and argument.

5. A point travels along $x = 3\cos t$, $y = 2\sin t$. Eliminate the parameter and describe one complete traversal.

6. Explain why a constant-speed circular trajectory is periodic but the spiral $z_n = (0.8e^{i\pi/3})^n z_0$ is not periodic when $z_0 \neq 0$.

7. Compute the finite sum $1 + i + i^2 + i^3$ and interpret its geometric cancellation.

8. Describe a real-world process that could be analyzed with at least three of the course's mathematical representations.